THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 8: **Heredity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which cross will always give all tall offspring? 1

(a) tt x tt **(b)** Tt x Tt **(c)** Tt x tt **(d)** TT x tt

2. A cross of RrYy with rryy gives offspring in the ratio 1

(a) 3 : 1 **(b)** 9 : 3 : 3 : 1 **(c)** 1 : 1 : 1 : 1 **(d)** 1 : 1

3. A woman with two X chromosomes passes 1

(a) one X to each child **(b)** a Y to sons

(c) both X to daughters **(d)** nothing

4. In which cross would 50% of the offspring be dwarf? 1

(a) TT x tt **(b)** TT x TT **(c)** Tt x tt **(d)** Tt x TT

5. In the F2 of a dihybrid cross between round yellow and wrinkled green parents, which combination is a new (non-parental) type? 1

(a) round green **(b)** round yellow

(c) wrinkled green **(d)** all of these are parental

6. A man passes his X chromosome to 1

(a) all his daughters **(b)** half of his daughters

(c) all his sons **(d)** half of his sons

7. Free earlobe (F) is dominant over attached (f). A child with attached earlobes can be born to parents with 1

(a) both heterozygous free earlobes **(b)** both homozygous free earlobes

(c) FF x Ff **(d)** FF x ff

8. Mendel's experiments disproved the idea that 1

(a) traits of parents blend in offspring **(b)** pea plants have flowers

(c) characters can be inherited **(d)** plants have traits

9. A man passes his Y chromosome to 1

(a) half of his daughters **(b)** all his sons

(c) half of his sons **(d)** all his daughters

10. Which of the following is an example of a character controlled by a pair of genes? 1

(a) Language spoken **(b)** A haircut

(c) A scar **(d)** Seed shape in peas

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why can a trait disappear in one generation and reappear in the next? 2

12. Out of 400 F2 plants of a monohybrid cross, how many are expected to be homozygous? 2

13. Distinguish between a homozygous and a heterozygous tall plant by a test cross. 2

14. A sperm has 22 + Y. What is the sex of the child it fertilises? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A woman who has XX gets married to a man with XY. Using a flow chart, state the possible sex of children and the ratio. 3

16. If both parents have the heterozygous genotype Rr for a trait, what is the probability that a child is rr? Explain. 3

17. Why do the two daughters of the same parents not look exactly alike? 3

OR

How does Mendel's experiment explain that factors for a trait exist as pairs and separate during gamete formation? 3

18. Explain how DNA gets copied and passed on, and why copying errors matter to heredity. 3

OR

Explain why a recessive trait can be present in a person who does not show it. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Differentiate between acquired and inherited traits with examples. Explain with an example how a trait acquired by an individual does not get inherited, and why a trait inherited by the sperm may be expressed in the offspring. 5

OR

Using a cross, prove that the F2 of a dihybrid cross contains 4 types of phenotypes and 9 types of genotypes. 5

20. Mendel crossed tall pea plants with dwarf plants and recorded the F1 and F2. Explain how he concluded that the factors are particulate and not blending. 5

OR

A pea plant with round yellow seeds is crossed with a plant having wrinkled green seeds and all offspring are round yellow. Give the genotypes and explain. A second cross gives half round yellow and half round green. Give the genotypes. 5

END OF QUESTION PAPER — Total: 40 Marks